

Solving One- & Two-Step Equations

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Use inverse operations to isolate the variable, doing the same to both sides. One-step: $x + 7 = 12 \rightarrow x = 5$. Two-step: $2x + 3 = 11 \rightarrow 2x = 8 \rightarrow x = 4$. Undo add/subtract first, then multiply/divide.

Round 1 — Retrieve It

1. What does it mean to "isolate the variable"?

2. Why must you do the same operation to both sides of an equation?

3. Solve the one-step equation: $x + 7 = 12$.

4. In a two-step equation, which do you undo first: add/subtract or multiply/divide?

Round 2 — Sketch It

Draw a balance scale for $2x + 3 = 11$. Show the first step (remove 3 from both sides) and the second step (split both sides into 2 equal groups). Circle the value of one x .

Solving One- & Two-Step Equations — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Solve: $x - 4 = 9$.

2. Solve: $3x = 21$.

3. Solve: $5x - 2 = 18$.

Round 4 — Reason from Evidence

A student solved $4x + 5 = 25$ by first dividing every term by 4. Explain why undoing the +5 first is easier, then solve the equation correctly showing both steps.

Solving One- & Two-Step Equations — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the order of steps you use to solve a two-step equation, and why that order works. Use one example that is not printed on the cue card.

C**Connect**

Connect solving an equation to a real situation, such as saving money. Write a word problem that matches $2x + 3 = 11$ and say what x stands for.

E**Extend**

Create your own two-step equation with a whole-number answer. Show every step and check your answer by substituting it back in.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Solving One- & Two-Step Equations

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Get the variable by itself on one side of the equation.
2. To keep the equation balanced (both sides stay equal).
3. $x + 7 = 12 \rightarrow x = 12 - 7 = 5$.
4. Undo add/subtract first, then multiply/divide.

Sketch It — look for

Balance shows $2x + 3 = 11$, then $2x = 8$ after removing 3 from both sides, then $x = 4$ after splitting each side into 2 equal groups.

Apply It

1. $x - 4 = 9 \rightarrow x = 9 + 4 = 13$.
2. $3x = 21 \rightarrow x = 21 \div 3 = 7$.
3. $5x - 2 = 18 \rightarrow 5x = 20 \rightarrow x = 4$.

Reason from Evidence — look for

Undoing +5 first avoids fractions. $4x + 5 = 25 \rightarrow 4x = 20 \rightarrow x = 5$. (Dividing first gives $x + 1.25 = 6.25$, which still works but is messier.)

ACE — Extend look-for

Many answers possible. Check that the equation is two-step, the steps are correct, and the check by substitution is shown.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.